Assignment 7- Solutions

MATH 3042- Assignment 7 Full Name:
Set:

1. A cybersecurity analyst studies the **proportion of login attempts that fail due to incorrect passwords**.

From a sample of **400 login attempts**, a confidence interval for the true proportion of failed attempts is:

$$(0.27\,,0.33)$$

$n = 400$

$\hat{p} = \frac{0.27+0.33}{2} = 0.3$

$E = \frac{0.33-0.27}{2} = 0.03$

Answer the following by using manual calculations:

a) What is the **sample proportion** of failed attempts? 30%

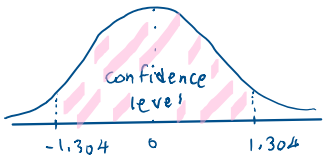
b) What is the **margin of error**? 3%

c) What is the **standard error of the proportion**? $SE = \sqrt{\frac{\hat{p}\hat{q}}{n}} = \sqrt{\frac{0.3\times0.7}{400}} = 0.023$

d) How many login attempts in the sample were **failures**? $n\hat{p} = (400)(0.3) = 120$ failed attempts

e) What is the confidence level?

$$z_{\frac{\alpha}{2}} = \frac{0.03}{0.023} = 1.304$$



confidence level = pnorm(1.304) − pnorm(−1.304)

$= 0.80957 \simeq 81\%$

2. A mobile app company wants to estimate the **mean load time** of its application.

Previous data suggest the **standard deviation is 2.5 seconds**. $\sigma = 2.5\,s$ (given)

The company wants the estimate to have a **margin of error of at most 0.5 seconds** with **95% confidence**.

1. What minimum **sample size** is required?
2. If the company instead wants the margin of error to be **0.2 seconds**, what sample size is required?

$E = 0.5\,s$ $\alpha = 5\%$

$Z_{\frac{\alpha}{2}} = 1.95996$

1. $$n = \left(\frac{1.95996 \times 2.5}{0.5}\right)^2$$

$$n = 96.04 \simeq \boxed{97}$$

2. $$n = \left(\frac{1.95996 \times 2.5}{0.2}\right)^2 = 600.2 \simeq \boxed{601}$$

3. A developer profiles memory usage (MB) for 25 runs of a new algorithm. From the sample, mean is 515 MB and standard deviation is 18 MB.
   a) Construct a 95% CI for the true mean memory usage.
   b) Write the statistically correct and the loosely correct interpretations of the CI above.
   c) Is 512 MB plausible as the true mean? Explain.
   d) How does the CI change if you use 90% confidence instead?

$n = 25 \quad \bar{x} = 515 \quad S = 18 \quad t_{\frac{\alpha}{2}, 24} = \text{abs}\left(\text{qt}\left(\frac{0.05}{2}, 24\right)\right) = 2.063899$

$SE = \frac{S}{\sqrt{n}} = \frac{18}{\sqrt{25}} = 3.6 \qquad E = (2.063899)(3.6) = 7.43 \text{ MB}$

$\bar{x} \pm E \quad \rightarrow UCL = 515 + 7.43 = 522.43 \text{ MB}$

$\quad\quad\;\; \rightarrow LCL = 515 - 7.43 = 507.57 \text{ MB}$

a) 95% CI for μ: $(507.6, 522.4)$ MB

b) If we repeatedly obtain samples of size 25 and construct CI for each, 95% of the constructed CI's will contain μ. Loosely speaking, we are 95% confident that μ is between 507.6 and 522.4 MB

c) Yes, 512 is in the CI. It is a plausible value for μ.

d) Since $t_{\frac{0.10}{2}, 24} = 1.711$, we will obtain a smaller margin of error $(E = 6.16)$ and a narrower / tighter confidenc interval:

$(508.8, 521.16)$ MB

4. A network engineer captures 2,000 packets and finds 38 are lost.

$n = 2000$ $\hat{p} = \frac{38}{2000} = 1.9\% = 0.019$

a) Check the conditions for the Wald CI.

b) Construct a 95% CI using prop.test().

c) Construct the Wald CI manually and compare.

d) Management claims the loss rate is < 3%. Does your CI support this?

a) $n\hat{p} = 38 \geq 10$ Yes

$n\hat{q} = 1962 \geq 10$ Yes

b)

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> prop.test(38,2000,0.95)

        1-sample proportions test with continuity correction

data:  38 out of 2000, null probability 0.95
X-squared = 36476, df = 1, p-value < 2.2e-16
alternative hypothesis: true p is not equal to 0.95
95 percent confidence interval:
 0.01366244 0.02625785
sample estimates:
    p
0.019
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95% wilson CI for π:

$\boxed{1.37\% < \pi < 2.63\%}$

c) $\hat{p} = 0.019$

$\hat{q} = 1 - 0.019 = 0.981$

$SE = \sqrt{\frac{0.019 \times 0.981}{2000}} = 0.003053$ $Z_{\frac{\alpha}{2}} = 1.95996$

$E = (1.95996)(0.003053) = 0.005983$

$\hat{p} \pm E =$ UCL $= 0.024983 \simeq 2.5\%$, LCL $= 0.013017 \simeq 1.3\%$

$\boxed{1.30\% < \pi < 2.5\%}$ 95% wald CI for π

Comparison: very close, basically the same.

d) Yes, all values in both CI's are less than 3%